

Detection of Early Blight and Septoria Leaf Spot from Tomato Leaf Image using Customized CNN Model with Atrous Convolution Layer

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Abstract—Deep learning (DL) based smart detection of plant disease using mobile images can revolutionize agricultural sector. In this article, we present a customized DL models, designed with 3 Convolutional Neural Network layers and 15 additional custom layers of Atrous Convolution for detecting two most common and severe diseases of tomato plant: early blight and septoria leaf spot from the leaf image. The customized model with dilation rate (DR) 32 can achieve accuracy, precision and recall 98.1%, 98.1% and 98.1%, respectively. The custom layer and DR are optimized through a search in terms of accuracy, memory footprint and parameters. While comparing with other standard models, the proposed model provides most efficient performance with only 18 layers, least execution time (only 0.435 s), and 2.68 MB memory footprint. The proposed model enables quick assessment with limited facility device. With proposed model early stage of diseases in tomato farm would allow farmers to take actions in the right time limiting the spread of infection with minimal resources.

Index Terms—Tomato leaf, Septorial Leaf spot, Early blight, Custom layers, Deep learning, Atrous Convolution.

I. INTRODUCTION

As per the report of Ministry of Agriculture and Farmers Welfare, approximate 19% of Indian GDP are contributed by the agricultural sector [1] with 46% of India's workforce to be directly related with the sector [2]. Yet, recent reports mention that in India, yearly crop lost due to various diseases and pest attacks is almost 18% [3] which causes huge economic losses and farmer's suffering.

The waste of agricultural output may even goes upto 50% for some specific crops and vegetables, like tomato [4]. With a notable production of 20.62 million metric tonnes in the fiscal year 2022, tomatoes are a staple food in India [5]. India is the world's second-largest producer of tomatoes, and its varied agroclimatic conditions allow for year-round tomato cultivation [4]. Yet high rate of pest attacks in Tomato farm causes low quality yield, yield loss, environmental threat, inflation and lower nutrition to consumers [6].

In case of tomatoes, bacteria or fungi or poor cultivation habits may cause 16 diseases while insects introduce

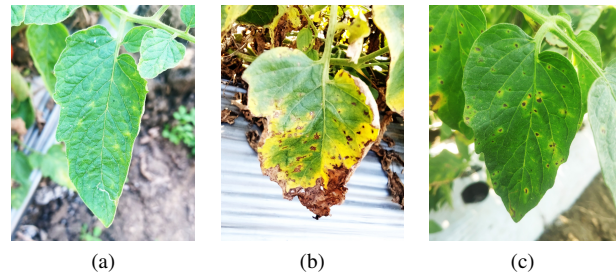


Fig. 1: Mobile camera photo of (a) a healthy leaf, (b) leaf affected with EB symptom, (c) leaf affected with SLS (taken by Google Pixel 8A, from Jalaram Tomato Farm, Durg, India).

5 other types of diseases. Among all, the most common and economically severe diseases are Early blight (EB) and Septoria leaf spot (SLS) [7]. The epidemic of SLS, caused by *Septoria lycopersici*, reduces fruit quality and yield even by 50% [7]. Plants infected by SLS display profound changes in photosynthesis, photorespiration, and defense responses [8]. The attack is more severe in tropical regions like India and Brazil where temperatures ranges from 20°C to 25°C and have high humidity [8], [9]. On the other hand, *Alternaria solani*, the causative agent of the EB disease of tomato, spreads indomitably under frequent rainfalls, high humidity, and medium-to-high temperatures: a common weather of India. Under severe infections, EB may cause up to 80% yield loss [7]. Owing to highest severity, in this work we focus on digital identification of EB and SLS infected leaves and healthy leaves based on mobile image. (viz. Fig. 3)

The endemic of tomato disease can be avoided by early detection of infection. However, such detection requires expertise that a common farmer lacks. An app based digital detection of the infection using mobile image data could be a smart, quick and inexpensive solution to this challenge and may safeguard the production bridging the gap and saving the sector. In recent literatures, several machine learning (ML) and deep learning (DL) based image processing model have been reported for tomato disease detection by extracting features from leaves' photo [4], [10].

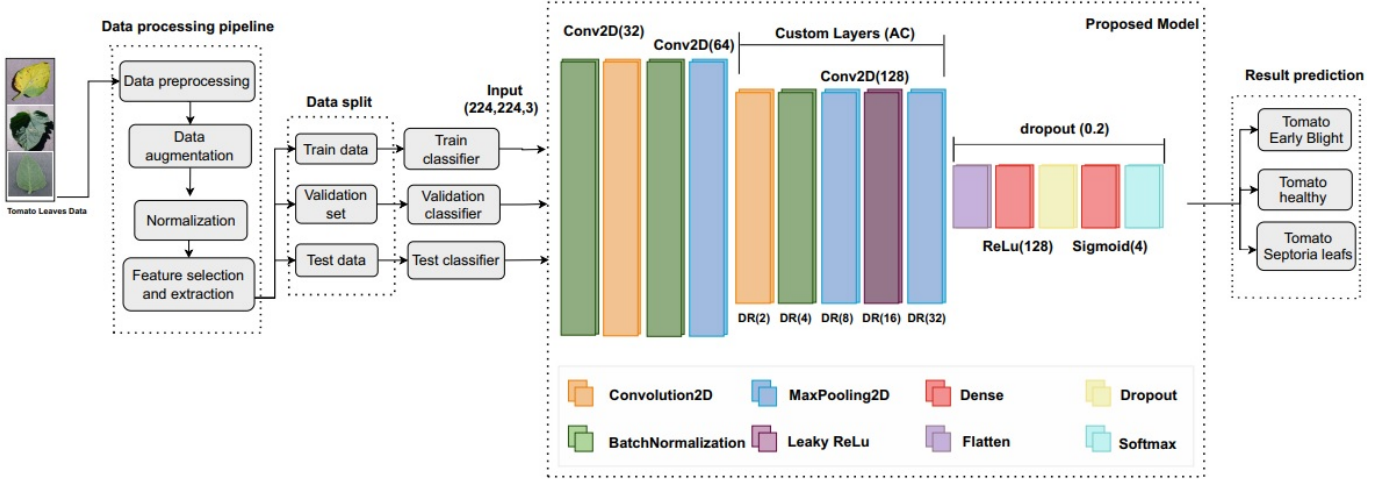


Fig. 2: Architecture of the proposed composite DL model indicating convolution layers and DR for three-class classification

Support vector machine has been used one of the common classifier to detect EB and bacterial spot in a few works [11]–[13] but that fails to handle the variance on image data. To handle image data most popular approach is to invoke convolutional neural network (CNN). Initial work using CNN were mostly on pre-trained DL models like LeNet [14], AlexNet [15], Inception V3, VGG-16, VGG-19 [16], ResNet [?] and choosing the most efficient one among them. Such approach limit the accuracy only to a moderate level and often based on a small dataset [15], [17].

Higher accuracy is achieved with hybrid models [18] with transfer learning [19] or introducing the Inception-v1 module, Inception-ResNet-v2 module [20], Nevertheless, such blending makes the model complicated thus optimization in terms of epoch, learning rate, training time and other parameters are now essential [6]. A comparative study on optimizer is given in [21], [22]. Customized models by introducing max pooling layers achieved an accuracy upto 91.2% [23]

Despite several approaches, not much higher accuracy is achieved while detecting SLS. Actually, it is found the detection accuracy in case of tomato leaves is found much less than the others like cotton or grape [19]. Apart from limited accuracy, the number of parameters, custom layers and memory footprints are also need to customized to implement the concept within the frame work of low cost mobile so that common farmers can afford. In this work, we focus on detection of both EB and SLS for tomato and develop a customized CNN model with minimum memory fruit print, execution time, custom layers ensuring higher accuracy.

II. THE IMAGE DATA FOR HEALTHY AND SICK LEAVES

All data for developing and training a suitable model for the detection of EB and SLS are collected from Kaggle data source [24]. To ensure consistency and to improve the model performance all images are resized to a fixed dimension of 224×224 pixels with three color channels (RGB). This resolution

helps balance computational efficiency and detail retention, ensuring that critical features of the leaf images are preserved.

To address data imbalance and to improve the model's generalization, data augmentation is adopted. Augmentation techniques such as random rotation, flipping, scaling, and translation were used to increase the diversity of the dataset. The augmented dataset contains a total of 6988 images, with the following distribution: 2407 healthy, 2400 EB infected, and 2181 SLS infected leaf images.

III. A CUSTOMIZED CNN MODEL FOR THREE CLASS CLASSIFICATION

This section describes the architecture of the CNN model. At first a simple CNN framework is adopted with following parameters:

- Convolution Layers: 3 (Simple CNN)
- Dropout Rate: 0.2
- Max Pooling Layers: 7
- Activation Function: ReLU, Softmax
- Epochs: 20
- Learning Rate: 0.001

With the large size of the training dataset we could go for a lower dropout rate assuring faster convergence. Convergence is also fastened by the sparsity introduced by Rectified Linear Unit (Relu) [25]. Seven 2×2 max pooling layers are used to reduce noise which is a common yet severe concern in case of agricultural data, specially when captured by common farmer with low cost mobile camera. Learning rate is automatically set by Adam algorithm. As we studied the variation of model accuracy with Epoch we found 20 is an optimal number. The value of epoch is much lower than the earlier work [19]. This lower value of epoch reduces the probability of overfitting and counterbalance the limitation of small dropout rate.

In the very next step, we compare the performance of the proposed model with other standard pre-trained DL model, namely, ResNet50, InceptionV3, VGG16, MobileNetV2, and DenseNet121. The performance were compared on the same

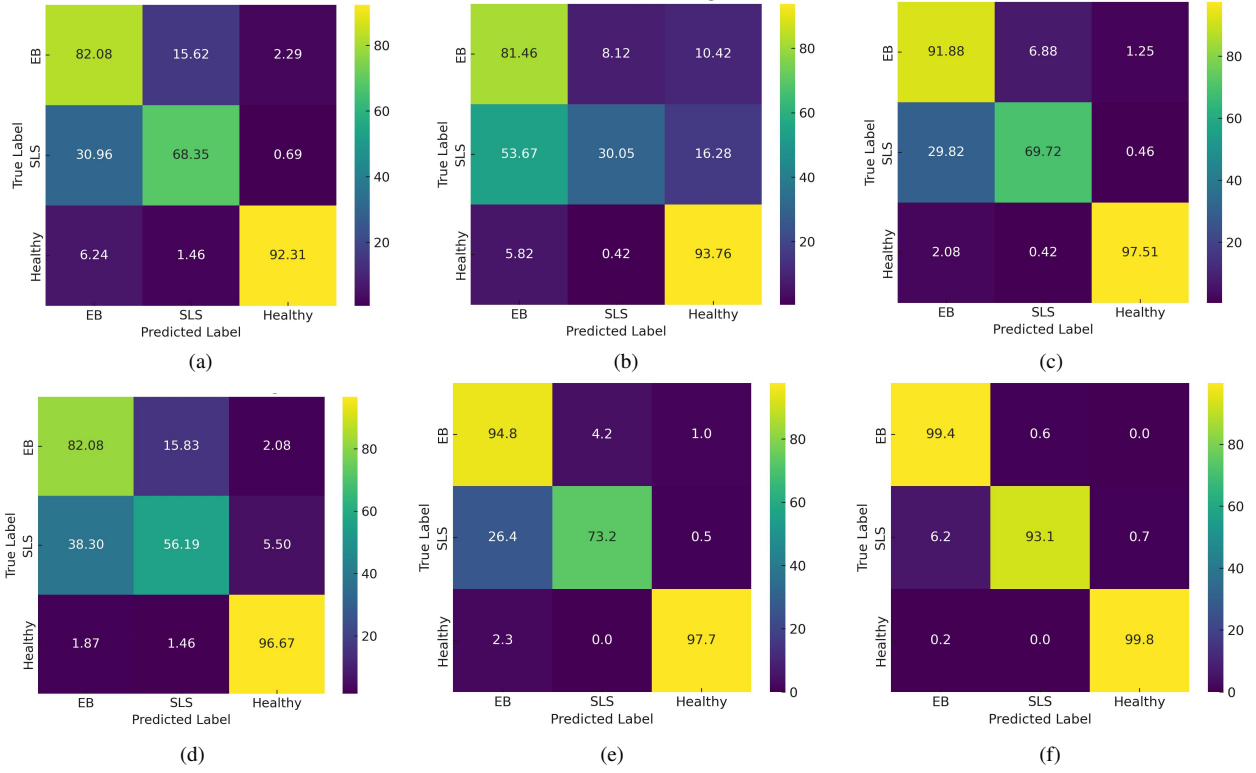


Fig. 3: Confusion matrices for 3 class classifications for tomato leaf disease using 6 different models: (a) VGG16, (b) RESNET50, (c) MobileNetV2, (d) INCEPTIONV3, (e) DENSENet121, and (f) the proposed CNN-AC composite models.

data set and corresponding performance indices: Precision (Pre), Recall (Rec), Accuracy (Acc), F1 Score and Execution times summarized in Table I.

Interestingly, our simple CNN model achieved an accuracy of 90.03%, which is better than all other pretrained models. While this improvement is promising, it still fall short of the target accuracy of more than 95%. This finding highlighted the potential of the custom CNN model but also points out to the need for further refinements.

A. Customized CNN Model with Atrous Convolution Layers

To further enhance the performance of the simple CNN model, we introduced custom layers between simple CNN layers and the dropout block, viz. Fig. 2. We redesigned the basic CNN model with five different types of custom layers: (i) Atrous convolution (AC), (ii) Depthwise separable convolution, (iii) Inception modules, (iv) Spatial pyramid pooling and (v) Squeeze excitation blocks. Once again the performances of

TABLE I: Performance comparison between a simple CNN model (developed in this work) and other pretrained models

Model Names	Acc (%)	Pre (%)	Rec (%)	F1-score	Time (s)
DenseNet121	89.0	90.2	90.2	0.902	3.40
InceptionV3	79.0	79.1	79.1	0.791	3.13
MobileNetV2	86.9	87.9	86.9	0.869	3.63
ResNet50	69.6	69.6	69.6	0.696	5.79
VGG16	81.1	82.0	81.3	0.813	5.74
Simple CNN	90.0	90.4	90.3	0.904	2.69

TABLE II: comparison between different type of custom models (9 custom layer in addition to 3 basic CNN layers)

Type of custom layer	Acc (%)	Pre (%)	Rec (%)	F1-score	Time (s)
Atrous convolution (DR=8)	96.9	96.8	96.8	96.8	3.18
Depthwise separable conv.	93.9	94.0	93.9	93.9	1.62
Inception modules	93.5	93.5	93.5	93.5	0.01
Spatial pyramid pooling	94.3	94.3	94.3	94.3	0.01
Squeeze excitation blocks	95.7	95.6	95.6	95.6	17.42

these five different custom models are compared and presented in Table II. It is evident that out of all five custom models, only two models: with AC and with Squeeze Excitation block can achieve the target efficiency of more than 95%. Furthermore, out of those two custom model with AC layers have the much lesser execution time.

Consequently, we study further with AC as the most suitable type of custom layers. Atrous convolution is also known as dilated convolution, allows the convolutional filters to have a larger receptive field without increasing the number of parameters. This approach enabled the model to capture more intricate features in the data, leading to improved classification performance.

So, as a final case study towards proposing the custom model for detecting EB and SLS symptom in tomato leaf, we systematically train different configurations of AC-CNN custom model with different number of AC layers with varying dilation rates (DR). The objective of this case study is

TABLE III: Comparison between number on layers and increased dilation rate (DR) of AC layers

No of AC Layers	DR	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Total	Memory Foot-print (MB)	Time (s)
0	0	93.7	93.7	93.7	0.936	25710415	98.08	2.6
3	2	93.1	93.1	93.0	0.930	12939727	49.36	5.23
6	4	92.1	93.3	92.0	0.919	3454031	13.18	4.98
9	8	96.9	96.9	96.9	0.969	1193679	04.55	3.98
12	16	97.6	97.6	97.6	0.976	686415	02.62	2.82
15	32	98.1	98.1	98.1	0.981	703439	02.68	2.49
18	64	96.4	96.5	96.3	0.963	851535	03.25	2.48
21	128	95.4	97.5	97.4	0.974	999631	03.81	5.84

maximizing the accuracy and minimizing the execution time with a minimal burden on memory foot print. The results of these case study are summarized in Table III. The optimal configuration for the proposed model is found with 15 AC layers with a dilation rate (DR) of 32, resulting in an accuracy of 98.1% and an execution time of 2.49 second with memory footprint only 2.68 MB. Lower or higher the number of AC layers make the performance poorer.

In conclusion, the inclusion of 15 AC layers (5 blocks) with a dilation rate of 32 along with basic 3 CNN layers improved the model accuracy from 90% to 98.1%, and is presented here as the final proposed model for EB and SLS detection.

IV. RESULTS AND DISCUSSION

Once performance of the proposed DL model (CNN+AC) (Table III) are compared with other pre-trained models (Table I), the proposed models found to have almost 7% better accuracy than the second best one (DenseNet121). Corresponding confusion matrices, presented in Fig. 3, illustrates that the proposed model demonstrates a substantial reduction in misclassifications, particularly for the classes that are frequently

TABLE IV: Comparison of accuracy as achieved by different works for tomato disease detection

Technique	Model	Accuracy	Reported in
Deep Learning	VGG6	95.5%	[14]
AI Framework	Pixel Replace	92.3%	[13]
Deep Learning	CNN+VGG	95.7%	[19]
Bayesian Optimization	LeNet CNN	95.0%	[22]
K-mean Cluster	NN Classifier	92.5%	[17]
Feature Pyramid	YOLOv6	94.0%	[26]
Deep Learning	Custom CNN	95.0%	[27]
Transfer Learning	Xception, ResNetV2	NA	[28]
Residual Network	ResNet50-DPA	96.3%	[29]
Deep Learning	CNN + AC	98.1%	This work

confused by other models. This improvement is attributed to the enhanced feature extraction capability of AC layers, which allow better spatial information retention while keeping computational costs low.

To further analyze the model's performance, the receiver operating characteristic (ROC) curves and t-distributed stochastic neighbor embedding (t-SNE) plots are presented in Fig. 4. The ROC curves confirms that the proposed model maintains a high true positive rate while minimizing false positives. The t-SNE plots provide a two-dimensional visualization of the feature representations, showing clear clustering of the three main categories: EB, SLS, and healthy leaves.

Additionally, the performance of the proposed model (CNN+AC) is compared with other reported works on tomato disease detection in Table IV and is found to achieve maximum efficiency among all models compared.

V. CONCLUSION

This work presents a customized CNN model with 15 AC layers (DR = 32) for the detection of EB and SLS in tomato leaf in contrast to healthy leaf. The custom layers type and number are selected optimizing accuracy, memory footprint and execution time. Finally an accuracy of 98.1% is achieved.

The comparison of the proposed model with other reported works on tomato disease detection highlights its effectiveness: A significant increase in accuracy while maintaining relatively low computational cost. This model provides a promising solution for real-time disease detection using mobile devices with high precision and minimal misclassification, making it a valuable tool for precision agriculture. In future step, further disease classes will be included into the detection model with field level validation using mobile app.

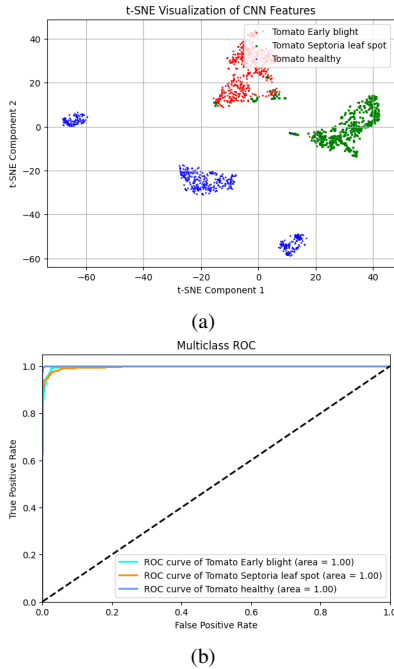


Fig. 4: (a) t-SNE and (b) ROC plot of the proposed model

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